Exp No:9. Experiment to understand Logistic Regression for a given data set.

Code:

```
import numpy as np

import pandas as pd


df = pd.read_csv('Social_Network_Ads.csv')


data = {

   'UserID': range(1, 21),

   'Gender': [

      'Female',

      'Male',

      'Female',

      'Male',

      'Male',

      'Female',

      'Male',

      'Female',

      'Male',

      'Female',

      'Male',

      'Female',

      'Male',
```

```
    'Female',
    'Male',
    'Female',
    'Male',
    'Female',
    'Male',
    'Female',
],
'Age': [
    21,
    29,
    34,
    40,
    27,
    32,
    46,
    25,
    37,
    43,
    30,
    48,
    23,
    35,
    41,
```

```
    28,
    34,
    26,
    45,
    33,
],
'EstimatedSalary': [
    28000,
    45000,
    52000,
    88000,
    36000,
    58000,
    105000,
    30000,
    75000,
    92000,
    52000,
    115000,
    27000,
    90000,
    98000,
    41000,
    82000,
```

```
        64000,
        108000,
        60000,
    ],
    'Purchased': [0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1],
}

df = pd.DataFrame(data)
print(df)
print(df.head())

features = df.iloc[:, [2, 3]].values
label = df.iloc[:, [4]].values
print(label)

from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

for i in range(1, 101):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )

    model = LogisticRegression()
```

```
    model.fit(x_train, y_train)
    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    if test_score > train_score:
        print(
            "Test:",
            test_score,
            "Train:",
            train_score,
            "Random state:",
            i,
        )

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)

finalModel = LogisticRegression()
finalModel.fit(x_train, y_train)

print(
    "Training score:",
    finalModel.score(x_train, y_train),
```

)

print("Testing score:", finalModel.score(x_test, y_test))

from sklearn.metrics import classification_report

print(classification_report(label, finalModel.predict(features)))

Output:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
array([[    19,  19000],
       [    35,  20000],
       [    26,  43000],
       [    27,  57000],
       [    19,  76000],
       [    27,  58000],
       [    27,  84000],
       [    32, 150000],
       [    25,  33000],
       [    35,  65000],
       [    26,  80000],
       [    26,  52000],
       [    20,  86000],
       [    32,  18000],
       [    18,  82000],
       [    29,  80000],
       [    47,  25000],
       [    45,  26000],
       [    46,  28000]
```

```
array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1,
       0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0,
       1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0,
       1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1,
       0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 1,
       1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1,
       0, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0,
       1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1,
       0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1,
       1, 1, 0, 1], dtype=int64)
```

Result:

Hence , this code has successfully executed and output is been verified.